THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 2 (Suggested Solution)

1. (**Bookwork**)

   (a) See *Theorem 4.26* (with the set $A = [a, b]$):
   f is continuous at $c \in [a, b]$ if and only if for every sequence $\{x_n\}$ in $[a, b]$ that converges to c, the sequence $\{f(x_n)\}$ converges to $f(c)$, that is,

   $$\lim_{n\to\infty} f(x_n) = f(c) \ \left(= f\big(\lim_{n\to\infty} x_n\big)\right).$$

   (b) See *Theorem 4.32*:

   Theorem 4.32 (Boundedness Theorem) If $f : [a, b] \longrightarrow \mathbb{R}$ is continuous on $[a, b]$, then f is bounded on $[a, b]$, that is, there exists $M > 0$ such that $|f(x)| \leq M$ for any $x \in [a, b]$.

   Proof. Assume to the contrary that f is not bounded on $[a, b]$. Then for any $M > 0$, there exists $x \in [a, b]$ such that $|f(x)| > M$ (here x may depend on M). This implies that for any $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ such that $|f(x_n)| > n$. Since the sequence $\{x_n\}$ is bounded, by the *Bolzano-Weierstrass Theorem*, there exists a subsequence $\{x_{n_k}\}$ converging to some number p. Since we have $a \leq x_{n_k} \leq b$ for any $k \in \mathbb{N}$, we know that $a \leq p \leq b$ and so $p \in [a, b]$. Now f is continuous at p and hence by the *Sequential Criterion for Continuity* (part (a)) the sequence $\{f(x_{n_k})\}$ converges to $f(p)$. Since $\{f(x_{n_k})\}$ is a convergent sequence, it must be bounded. But this is a contradiction since

   $$|f(x_{n_k})| > n_k \geq k \quad \text{for any } k \in \mathbb{N}.$$

   Hence f must be bounded on $[a, b]$.

2. (a) For any $k \in \mathbb{N}$, choose a sequence $\{x_n\}$ of irrational numbers converging to $\frac{1}{2^k}$ (the existence of such a sequence is guaranteed by the hint). By definition of f, we have $f(x_n) = 0$ for any $n \in \mathbb{N}$, and so $\lim_{n\to\infty} f(x_n) = 0$. However, $f(\frac{1}{2^k}) = \frac{1}{2^{k+1}} \neq 0$ and hence f cannot be continuous at $\frac{1}{2^k}$ for any $k \in \mathbb{N}$.

   (b) We want to show that for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in \mathbb{R}$,

   $$|x - 0| = |x| < \delta \implies |f(x) - f(0)| = |f(x)| < \epsilon.$$

   If $x = \dfrac{1}{2^k}$ for some $k \in \mathbb{N}$, then we have

   $$|f(x)| = \left|\frac{1}{2^{k+1}}\right| = \frac{|x|}{2}.$$

   If $x \neq \dfrac{1}{2^k}$ for any $k \in \mathbb{N}$, then we have $|f(x)| = 0$.

Thus, for any $\epsilon > 0$, we can choose $\delta = \epsilon > 0$ such that

$$|x| < \delta \implies |f(x) - f(0)| = |f(x)| = \begin{cases} \dfrac{|x|}{2} < \dfrac{\delta}{2} < \delta = \epsilon & \text{if } x = \dfrac{1}{2^k} \text{ for some } k \in \mathbb{N}, \\ 0 < \epsilon & \text{otherwise .} \end{cases}$$

(c) Disproof:

Consider the sequences $\{x_n\} = \{\frac{1}{2^n}\}$ and $\{y_n\} = \{\frac{1}{3^n}\}$. Then we have

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{2^n} = 0 \quad \text{and} \quad \lim_{n\to\infty} y_n = \lim_{n\to\infty} \frac{1}{3^n} = 0.$$

Moreover,

$$f(x_n) = \frac{1}{2^{n+1}} = \frac{x_n}{2} \quad \text{and} \quad f(y_n) = 0 \quad \text{for all } n \in \mathbb{N}.$$

It follows that

$$\lim_{n\to\infty} \frac{f(x_n) - f(0)}{x_n - 0} = \frac{1}{2} \neq 0 = \lim_{n\to\infty} \frac{f(y_n) - f(0)}{y_n - 0}.$$

Hence we have found two sequences $\{x_n\}$ and $\{y_n\}$ such that x_n, $y_n \neq 0$ for all $n \in \mathbb{N}$ and

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n = 0 \quad \text{but} \quad \lim_{n\to\infty} \frac{f(x_n) - f(0)}{x_n - 0} \neq \lim_{n\to\infty} \frac{f(y_n) - f(0)}{y_n - 0}.$$

We conclude that f is NOT differentiable at 0.

3. (**Bookwork**) See *Tutorial 5, Q.1(a)-(c)*:

(a) Suppose $L > 0$. Take $\epsilon = \frac{L}{2} > 0$. Then there exists $N \in \mathbb{N}$ such that for any $n \in \mathbb{N}$, if $n \geq N$, then

$$\left|\frac{a_n}{b_n} - L\right| < \frac{L}{2}, \quad \text{or equivalently,} \quad \frac{L}{2}\, b_n < a_n < \frac{3L}{2}\, b_n.$$

If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} \frac{L}{2}\, b_n$ converges by the comparison test, and hence $\displaystyle\sum_{n=1}^{\infty} b_n$ converges.

Conversely, if $\displaystyle\sum_{n=1}^{\infty} b_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} \frac{3L}{2}\, b_n$ converges and hence $\displaystyle\sum_{n=1}^{\infty} a_n$ converges by the comparison test.

(b) Suppose $L = 0$. Take $\epsilon = 1$. Then there exists $N \in \mathbb{N}$ such that for any $n \in \mathbb{N}$, if $n \geq N$, then

$$\left|\frac{a_n}{b_n}\right| < 1, \quad \text{or equivalently,} \quad a_n < b_n.$$

By the comparison test, if $\displaystyle\sum_{n=1}^{\infty} a_n$ diverges, then $\displaystyle\sum_{n=1}^{\infty} b_n$ diverges as well.

(c) Take $a_n = \dfrac{1}{n^2}$ and $b_n = \dfrac{1}{n}$ for any $n \in \mathbb{N}$. Then $\{a_n\}$ and $\{b_n\}$ are sequences of positive terms. Moreover, we have

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{1}{n} = 0, \quad \sum_{n=1}^{\infty} a_n \text{ converges but } \sum_{n=1}^{\infty} b_n \text{ diverges.}$$

(d) Proof:

First of all, we observe that

$$\lim_{n\to\infty}(nc_n - 1) = 0, \quad \text{or equivalently,} \quad \lim_{n\to\infty} nc_n = 1 \tag{$*$}$$

as the series $\displaystyle\sum_{n=1}^{\infty}(nc_n - 1)$ converges. Let $a_n = \dfrac{c_n^2}{\sqrt{n}}$ and $b_n = \dfrac{1}{n^{5/2}}$ for any $n \in \mathbb{N}$. Then $\{a_n\}$ and $\{b_n\}$ are sequences of positive terms. Moreover, we have

$$\lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{\frac{c_n^2}{\sqrt{n}}}{\frac{1}{n^{5/2}}} = \lim_{n\to\infty}\frac{c_n^2}{\sqrt{n}}\cdot n^{5/2} = \lim_{n\to\infty} n^2c_n^2 = 1 \quad \text{by } (*).$$

Since $\displaystyle\sum_{n=1}^{\infty} b_n$ converges (p-series with $p = \frac{5}{2}$) and $L = 1 > 0$, by (a) we conclude that $\displaystyle\sum_{n=1}^{\infty}\frac{c_n^2}{\sqrt{n}}$ converges.

4. (a) True

- Since $\left|(-1)^{n+1}\left(\dfrac{a^2+n}{n^2}\right)\right| = \dfrac{a^2+n}{n^2} \geq \dfrac{n}{n^2} = \dfrac{1}{n}$ (≥ 0) for all $n \in \mathbb{N}$, and since the harmonic series diverges, the Comparison Test implies that the given series does not converge absolutely.
- Since $\frac{a^2+n}{n^2} > 0$ for all $n \in \mathbb{N}$, the sequence $\left\{\frac{a^2+n}{n^2}\right\}$ is decreasing (since

$$\frac{a^2+n+1}{(n+1)^2} = \frac{a^2}{(n+1)^2} + \frac{1}{n+1} < \frac{a^2}{n^2} + \frac{1}{n} = \frac{a^2+n}{n^2} \quad \text{for any } n \in \mathbb{N}),$$

and $\lim\limits_{n\to\infty}\frac{a^2+n}{n^2} = \lim\limits_{n\to\infty}\left(\frac{a^2}{n^2} + \frac{1}{n}\right) = 0$, thus, by alternating series test, the given series converges for any $a \in \mathbb{R}$.

It follows that the given series converges conditionally for any $a \in \mathbb{R}$.

(b) False

Define $f : (0,1) \longrightarrow [0,1]$ by $f(x) = |\sin(2\pi x)|$. Then f is continuous on $(0,1)$ as it is the composition of the absolute value function and the sine function which are both continuous on $(0,1)$. It remains to show that f is surjective.

For any $y \in [0,1]$, if $y = 0$, then we can choose $x = \frac{1}{2}$. If $y = 1$, then we can choose $x = \frac{1}{4}$. If $0 < y < 1$, then we can apply the Intermediate Value Theorem to f on $[\frac{1}{4}, \frac{1}{2}]$ and conclude that there exists $c \in (\frac{1}{4}, \frac{1}{2})$ such that $f(c) = y$. It follows that f is surjective.

(c) True

Since f is bounded on $\mathbb{R}$, there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in \mathbb{R}$. Let $g(x) = f(x) - x$ for all $x \in \mathbb{R}$. Since f is continuous on $\mathbb{R}$, $g : \mathbb{R} \longrightarrow \mathbb{R}$ is also continuous on $\mathbb{R}$. If $f(-M) = -M$ or $f(M) = M$, then we are done. Otherwise, we must have $f(-M) > -M$ and $f(M) < M$. In other words, we have

$$g(-M) = f(-M) + M > 0 \quad \text{and} \quad g(M) = f(M) - M < 0.$$

We can now apply the Intermediate Value Theorem to g on $[-M, M]$ to conclude that there exists $c \in (-M, M)$ such that $g(c) = 0$, that is, $f(c) = c$.

(d) <u>True</u>

Define $g : [0, \frac{1}{2}] \longrightarrow \mathbb{R}$ by

$$g(x) = f\left(x + \frac{1}{2}\right) - f(x) - \frac{1}{2}(f(1) - f(0))$$

for any $x \in [0, \frac{1}{2}]$. Then g is continuous on $[0, \frac{1}{2}]$ (since f is continuous on $[0, 1]$), and we have

$$g(0) = f\left(\frac{1}{2}\right) - f(0) - \frac{1}{2}(f(1) - f(0)) \quad \text{and} \quad g\left(\frac{1}{2}\right) = f(1) - f\left(\frac{1}{2}\right) - \frac{1}{2}(f(1) - f(0)) = -g(0).$$

Now if $g(0) = 0$, then ($x_1 = 0$ and $x_2 = \frac{1}{2}$) or ($x_1 = \frac{1}{2}$ and $x_2 = 1$) will do the job.
If $g(0) \neq 0$, then $g(0)$ and $g(\frac{1}{2})$ will have opposite signs and since g is continuous on $[0, \frac{1}{2}]$, by the Intermediate Value Theorem, there exists $x_0 \in (0, \frac{1}{2})$ such that $g(x_0) = 0$, that is,

$$f\left(x_0 + \frac{1}{2}\right) - f(x_0) = \frac{1}{2}(f(1) - f(0)).$$

Thus in this case, we can take $x_1 = x_0$ and $x_2 = x_0 + \frac{1}{2}$.